Worksheet 21 · Indices IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Apply the laws of indices • Evaluate negative and fractional indices • Simplify expressions involving indices		
Grade: / 47	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Indices*

Question 1 (4 marks)

(a) Simplify $t^9 \div t^3$

.....
(1)

(b) Simplify $w^5 \times w^7$

.....
(1)

(c) Simplify $(5xy^2)^3$

.....
(2)

Question 2 (2 marks)

(a) Simplify fully $\frac{15k^4m^3}{5km^2}$

.....
(2)

Question 3 (3 marks)

(a) Simplify n^0

.....
(1)

(b) Simplify $(3x^2y^5)^3$

.....
(2)

Question 4 (2 marks)

Simplify fully $(16x^8y^6)^{\frac{1}{2}}$

(2)

Question 5 (4 marks)

(a) Simplify fully $(x^{12}y^8)^{\frac{3}{4}}$

.....
(2)

Given that $3^n = \frac{3^x}{9^y}$

(b) find an expression for n in terms of x and y.

$n =$ .....
(2)

Question 6 (2 marks)

(a) Simplify fully $\left(\frac{256x^{20}}{y^8}\right)^{-\frac{1}{4}}$

(2)

Question 7 (3 marks)

Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

.....
(3)

Question 8 (4 marks)

(a) Simplify $a^7 \times a^4$

.....
(1)

(a) Simplify $e^8 \div e^2$

.....
(1)

(a) Write down the value of m, given that $3^4 \times 3^5 = 3^m$

$m =$ .....
(1)

(b) Write down the value of n, given that $(5^3)^7 = 5^n$

$n =$ .....
(1)

(c) Find the value of p, given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

Question 9 (6 marks)

(a) Given that $a = 3^x$ and $b = 3^y$

express in terms of a or b or a and b,

(i) 3^{2x}

.....

(ii) 3^{x+4y}

.....

(iii) 3^{y-1}

.....
(3)

$a = 3^x$ and $b = 3^y$

$ab = 2187$

$a^2b = 177147$

(b) Work out the value of x and the value of y.
Show your working clearly.

$x =$ .....

$y =$ .....
(3)

Question 10 (2 marks)

Given that $\frac{w^5 \times w^n}{w^3} = w^{10}$

(c) work out the value of n.

$n =$ .....
(2)

Question 11 (2 marks)

Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

(b) work out the value of n.

$n =$ .....
(2)

Question 12 (4 marks)

(a) Write down the value of x^0

.....
(1)

Given that $2^{-3} \times 2^9 = 2^n$

(b) find the value of n

$n =$ .....
(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

(c) find the value of m

$m =$ .....
(2)

Question 13 (4 marks)

(a) Simplify $(3k^2)^4$

.....
(2)

(b) Simplify $(21m^4n) \div (3n^{-5})$

.....
(2)

Question 14 (2 marks)

Simplify $(64p^9q^{12})^{\frac{2}{3}}$

.....
(2)

Question 15 **(3 marks)**

Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

……